

Introduction to Mathematics and Optimization

January, 2022

PROBLEM 1

Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the function given by

$$f(x, y, z) = x^2 + y^2 + (z - 1)^2.$$

Furthermore, let

$$g_1(x, y, z) = z - x$$

$$g_2(x, y, z) = z + x$$

$$g_3(x, y, z) = z - y$$

$$g_4(x, y, z) = z + y$$

$$g_5(x, y, z) = -1 - z$$

be functions from $\mathbb{R}^3$ to $\mathbb{R}$ and define

$$C = \{v \in \mathbb{R}^3 \mid g_1(v) \leq 0, \quad g_2(v) \leq 0, \quad g_3(v) \leq 0, \quad g_4(v) \leq 0, \quad g_5(v) \leq 0\}$$

for $v = (x, y, z) \in \mathbb{R}^3$. In this problem we focus on the optimization problem

$$\begin{aligned} \min f(v) \\ v \in C. \end{aligned} \tag{1}$$

(a) Show that (1) is a convex optimization problem.

(b) The condition $(x, y, z) \in C$ means that

$$z \leq x, \quad z \leq -x, \quad z \leq y, \quad z \leq -y, \quad z \geq -1. \tag{2}$$

Show that from $z \leq x$ and $z \leq -x$ one can conclude that $z \leq 0$. Explain likewise why one can conclude that $-1 \leq x \leq 1$ and $-1 \leq y \leq 1$ based on the inequalities in (2).

(c) Show that C is compact. Explain why this implies that the optimization problem (1) can be solved.

(d) Is a solution to the optimization problem unique? If so, why?

(e) Write down the KKT conditions corresponding to (1) in the variables $x, y, z, \lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5$ and explain from these why $(-1, -1, -1)$ cannot be an optimal solution to (1).

(f) Show that $(0, 0, 0)$ is an optimal solution to (1). Do the KKT conditions corresponding to (1) have a unique solution?

(g) Give a geometric interpretation of the optimization problem (1).

PROBLEM 2

Let A denote the matrix

$$\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$$

and $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ the function given by

$$f(x, y) = x^2 + xy.$$

(a) Find an invertible matrix B such that

$$B^T A B = \begin{pmatrix} 2 & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}.$$

(b) Show that A is not positive semidefinite.

(c) Show that f is not a convex function.

(d) Find two vectors $u, v \in \mathbb{R}^2$ such that

$$\begin{aligned} u^T A u &> 0 & \text{and} \\ v^T A v &< 0 \end{aligned}$$

possibly using the matrix B from (a).

(e) Find the critical points of f and determine their types (local minimum/maximum, saddle point, etc.).

PROBLEM 3

Let

$$f(x) = a_2x^2 + a_1x + a_0 \quad (3)$$

denote a function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by the real numbers a_2, a_1, a_0 . For $a_2 > 0$, this is a familiar parabola opening upward.

- (a) Compute $f''(x)$. What must a_2 satisfy for $f(x)$ to be a convex function? What must a_2 satisfy for $f(x)$ to be strictly convex?
- (b) Show that if the graph of the function f defined in (3) passes through the points $(0, 0)$, $(1, 1)$, $(2, 3)$, and $(3, 5)$, then the system of equations

$$a_0 = 0$$

$$a_2 + a_1 + a_0 = 1$$

$$4a_2 + 2a_1 + a_0 = 3$$

$$9a_2 + 3a_1 + a_0 = 5$$

has a solution in a_2, a_1 , and a_0 . Show that this is impossible.

- (c) Find a best-fit parabola (in the least squares sense) through the points in part (b).
- (d) Find a number y such that there exists a parabola passing through the points $(0, 0)$, $(1, 1)$, $(2, 3)$, and $(3, y)$.